

WEEK-2 DATA MODELING, DAX & SALES DASHBOARD INTRODUCTION

Retail Sales Performance Dashboard

This project focuses on analyzing retail sales data using Power BI. The report includes data preparation, DAX measures, KPI cards, charts, slicers, interactive analysis, and a final dashboard.

STEP 1 – Define the Objective

The objective of this project is to transform raw retail sales data into an interactive Power BI dashboard.

The dashboard is designed to analyze:

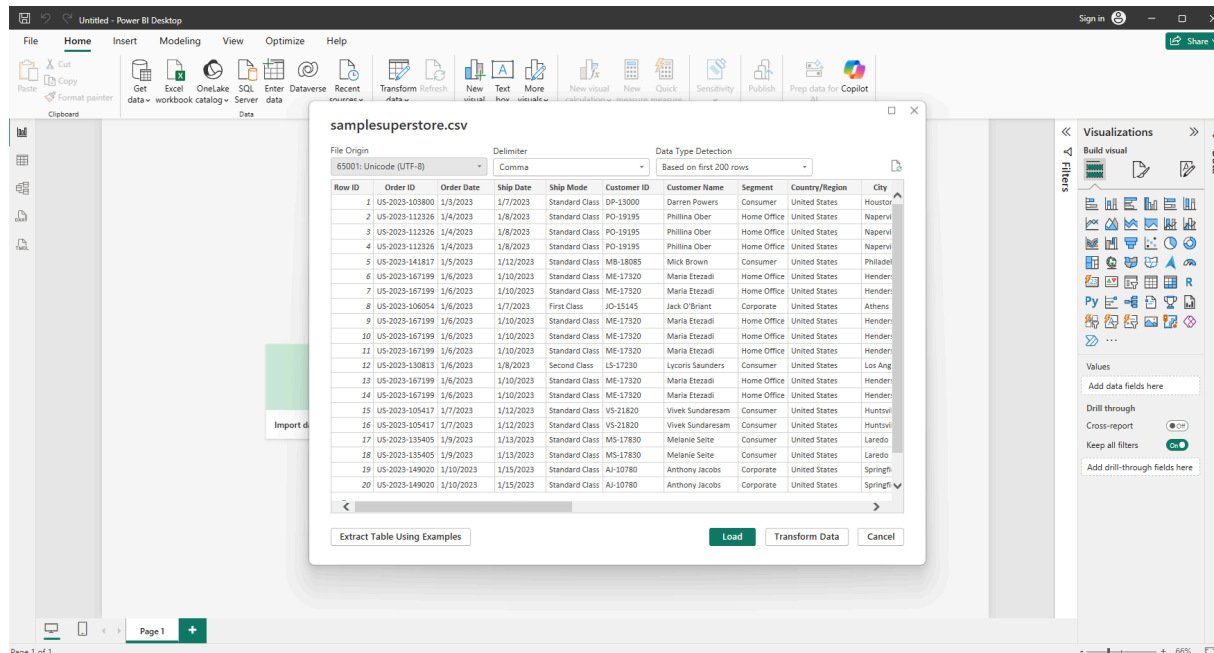
- Overall sales and profit performance
- Total orders and quantity sold
- Category-wise sales performance
- Sales trends over time
- Regional and segment-wise performance
- Interactive filtering using slicers

STEP 2 – Load the Dataset

The **SampleSuperstore** dataset was imported into Power BI Desktop.

The dataset contains retail transaction details required for sales, profit, order, quantity, category, region, segment, and time-based analysis.

Screenshot:



STEP 3 – Review the Dataset

The imported dataset was reviewed to understand the available fields and their purpose.

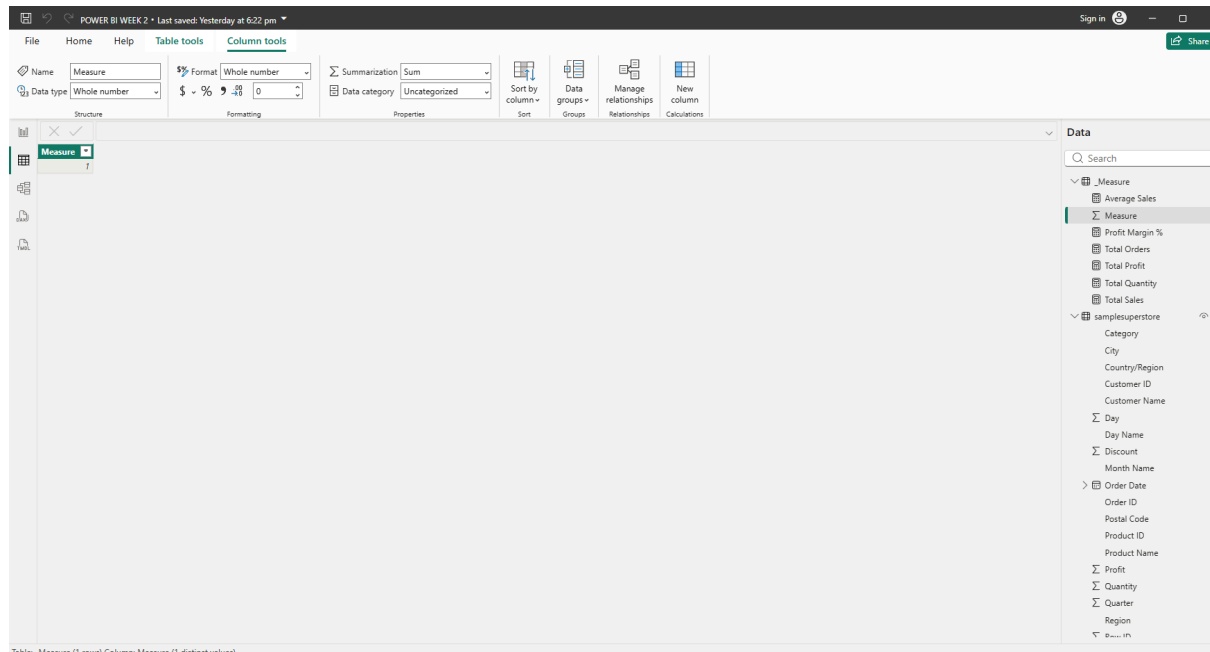
Important fields used in the project include:

- Category
- Sub-Category
- Region
- Segment
- Order Date
- Sales
- Profit
- Quantity
- Order ID

STEP 4 – Create the Measures Table

A separate **_Measures** table was created to store all the DAX measures used in the report.

This helps keep the calculations organized and makes the Power BI model easier to manage.



STEP 5 – Create Total Sales Measure

A DAX measure was created to calculate the total sales value.

Total Sales = SUM(SampleSuperstore[Sales])

This measure is used in KPI cards and other sales-related visuals.

STEP 6 – Create Total Profit Measure

A DAX measure was created to calculate the total profit generated from the sales data.

Total Profit = SUM(SampleSuperstore[Profit])

This measure is used to analyze the overall profitability of the business.

STEP 7 – Create Total Orders Measure

The Total Orders measure was created using the unique Order ID values.

Total Orders = `DISTINCTCOUNT(SampleSuperstore[Order ID])`

This measure represents the total number of unique orders.

STEP 8 – Create Total Quantity Measure

A measure was created to calculate the total quantity of products sold.

Total Quantity = `SUM(SampleSuperstore[Quantity])`

This measure helps analyze the total number of units sold.

STEP 9 – Create Average Sales Measure

The Average Sales measure was created to calculate the average sales value.

Average Sales = `AVERAGE(SampleSuperstore[Sales])`

This measure provides the average sales value for the records in the dataset.

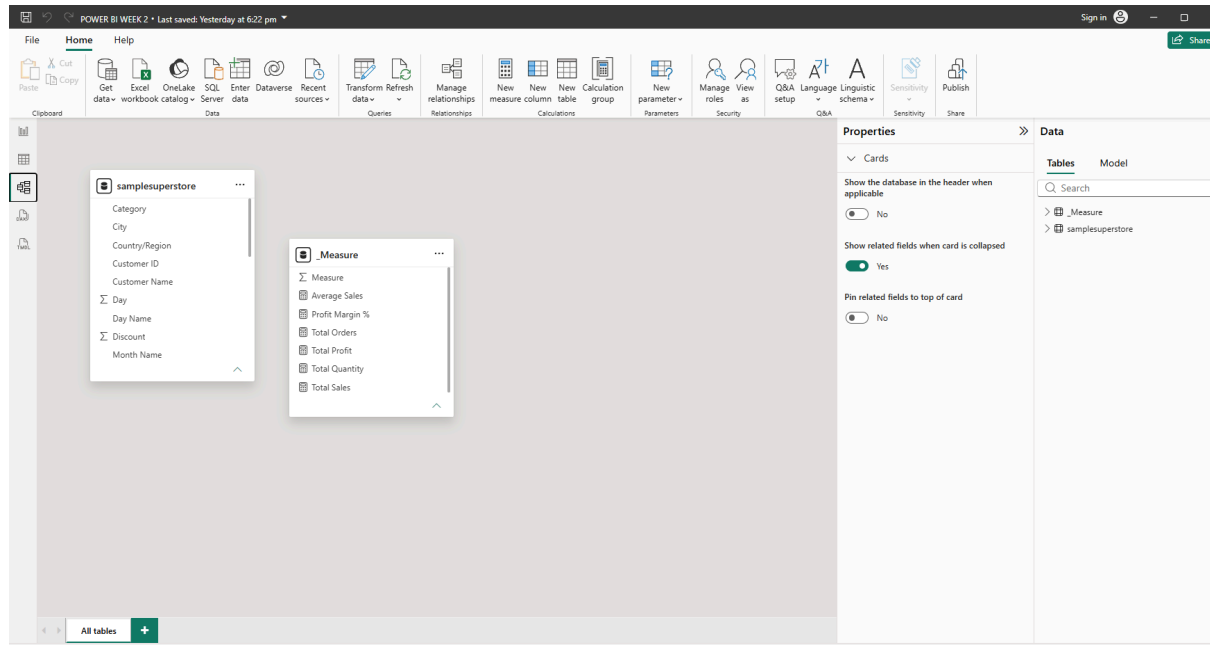
STEP 10 – Create Profit Margin % Measure

The Profit Margin % measure was created to calculate profit as a percentage of total sales.

Profit Margin % = `DIVIDE([Total Profit], [Total Sales], 0)`

This measure helps understand the relationship between sales and profit.

Screenshot:



Power BI Data Model showing the DAX measures created for Total Sales, Total Profit, Total Orders, Total Quantity, Average Sales, and Profit Margin %.

STEP 11 – Create KPI Cards

Six KPI card visuals were created to display the major performance indicators.

The cards display:

- **Total Sales**
- **Total Profit**
- **Total Orders**
- **Total Quantity**
- **Average Sales**

- **Profit Margin %**

These cards provide a quick overview of the overall business performance.

STEP 12 – Create Sales by Category Chart

A chart was created to compare **Total Sales across different Categories**.

The main categories analyzed are:

- Furniture
- Office Supplies
- Technology

This visual helps compare sales performance across product categories.

STEP 13 – Create Sales Trend Chart

A line chart was created to analyze **Total Sales over time**.

The year/date field was used on the axis and Total Sales was used as the value.

This visual helps identify changes and trends in sales performance over time.

STEP 14 – Create Category and Sub-Category Analysis

Category and Sub-Category fields were used to provide a more detailed view of product performance.

This allows the analysis to move from the overall category level to individual sub-categories.

STEP 15 – Add Slicers

Interactive slicers were added to allow users to filter the dashboard dynamically.

The following slicers were used:

- **Region**
- **Segment**
- **Year**
- **Category**

When a value is selected, the related visuals update according to the selected filter.

STEP 16 – Create Decomposition Tree

A **Decomposition Tree** was created for interactive sales analysis.

The visual was configured using:

- **Analyze:** Total Sales
- **Explain By:** Category

The Decomposition Tree allows Total Sales to be broken down into individual categories for detailed analysis.

STEP 17 – Apply Cross-Filtering

Cross-filtering was used to make the dashboard interactive.

When a category, region, segment, or other data point is selected, the related visuals are filtered according to the selection.

This improves the interactive analysis of the dashboard.

STEP 18 – Apply Interactive Tooltips

Interactive tooltips were used to display additional information when the user moves the cursor over a visual.

This provides more details without adding extra visuals to the dashboard.

STEP 19 – Apply Drill-Down Analysis

Drill-down functionality was used to move from a higher-level category to a more detailed sub-category level.

The analysis can therefore move from:

Category → Sub-Category

This provides a more detailed understanding of product-level performance.

STEP 20 – Build Page 1: Sales Overview

The first dashboard page was designed as a **Sales Overview** page.

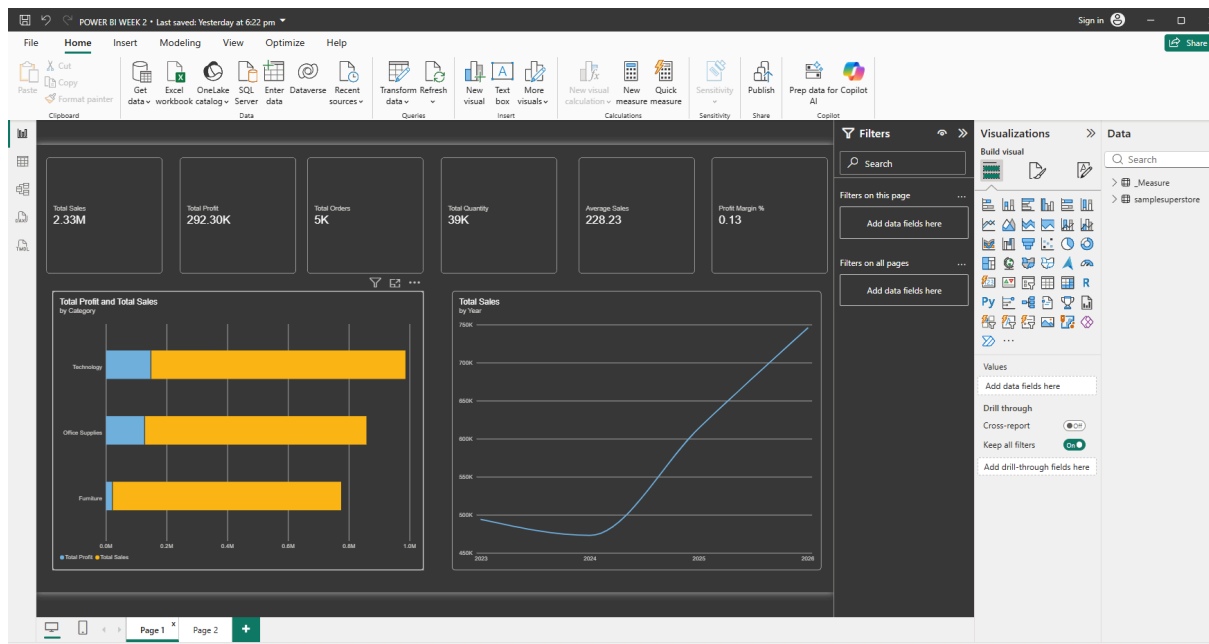
It contains:

- KPI Cards
- Sales by Category chart

- Sales Trend line chart

The page provides a quick summary of the overall sales and profit performance.

Screenshot:



STEP 21 – Build Page 2: Category Deep-Dive

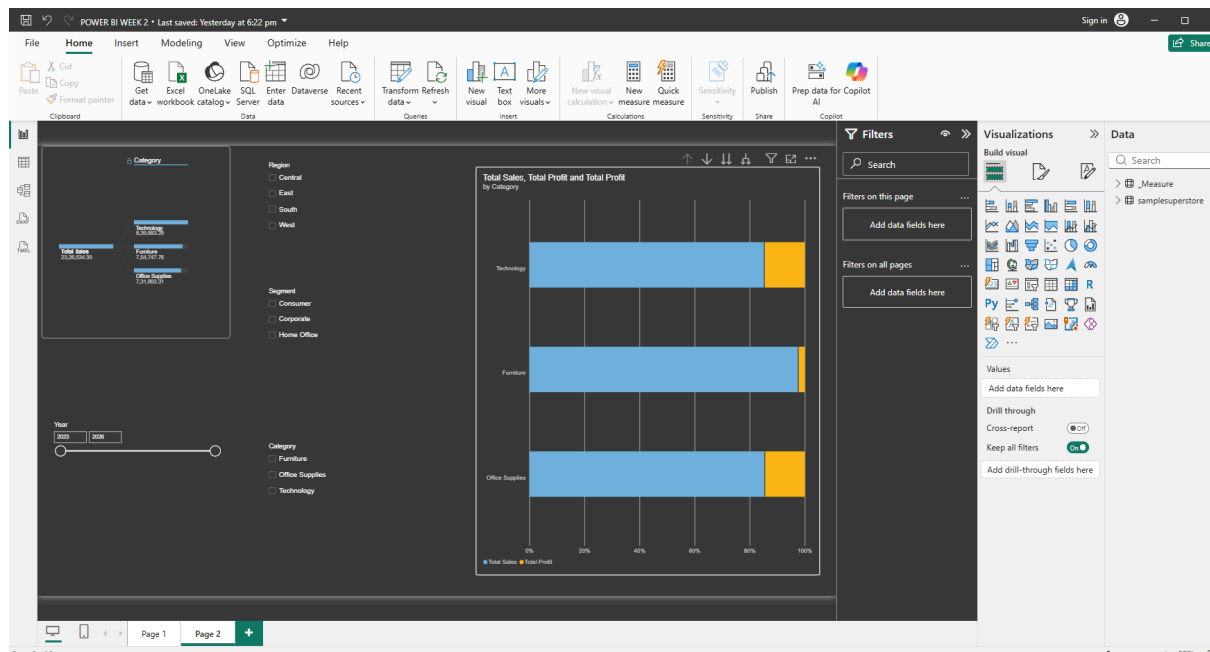
The second dashboard page was designed for detailed and interactive analysis.

It contains:

- Decomposition Tree
- Region slicer
- Segment slicer
- Year slicer
- Category slicer
- Category/Sub-Category analysis

This page allows users to explore the sales data using different filters.

Screenshot:



STEP 22 – Apply Dashboard Design

The dashboard visuals were arranged in a clear and structured layout.

The design includes:

- KPI cards at the top
- Charts arranged for easy comparison
- Slicers for interactive filtering
- Consistent formatting across the report
- Clear visual separation between summary and detailed analysis

STEP 23 – Test Interactivity

The dashboard was tested using different slicer selections and visual interactions.

The purpose was to verify that the charts, KPI cards, and other visuals respond correctly to user selections.

STEP 24 – Identify Key Insights

The dashboard provides several useful observations:

- Overall sales and profit can be viewed through the KPI cards.
- Category-wise sales performance can be compared using the charts.
- Sales trends can be analyzed over time.
- Region, Segment, Year, and Category filters provide interactive analysis.
- The Decomposition Tree provides a detailed breakdown of Total Sales.

STEP 25 – Conclusion

The Retail Sales Performance Dashboard was successfully developed using Power BI.

The project demonstrates the use of **data preparation, DAX measures, KPI cards, charts, slicers, Decomposition Tree, drill-down, tooltips, and cross-filtering** to create an interactive sales analysis dashboard.

The final dashboard provides a clear and structured way to analyze **sales, profit, orders, quantity, categories, and trends** from the SampleSuperstore dataset.